
Online Examination Report

Date:	17.07.2014
Attempts:	2
Examinee:	Ariho, William
Date of birth:	4/22/1989
Subject:	030-Flight Performance&Planing
Result:	14,59 %
Time used:	00:30:00 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	377	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
2	378	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	1,00	1,00
3	381	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
4	374	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
5	382	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	1,00	1,00
6	390	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
7	384	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
8	380	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	1,00	1,00
9	420	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
10	361	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
11	379	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
12	373	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	1,00	1,00
13	315	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
14	183	31.2	LOADING	0,00	1,00
15	257	31.3.1	Definition of centre of gravity	1,00	1,00
16	411	32.	PERFORMANCE - AEROPLANES	1,00	1,00
17	424	32.	PERFORMANCE - AEROPLANES	0,00	1,00
18	423	32.	PERFORMANCE - AEROPLANES	0,00	1,00
19	118	32.2.2	Effect of Variables on Single-Engine Aeroplane Performance	0,00	1,00
20	36	32.2.5.4	Landing	0,00	1,00
21	179	32.3.3.1	Take-off	0,00	1,00
22	34	32.4.1.7	Take-off climb	0,00	1,00
23	103	32.4.1.7	Take-off climb	0,00	1,00
24	189	32.4.1.7	Take-off climb	0,00	1,00
25	197	32.4.1.7	Take-off climb	0,00	1,00
26	47	32.4.3	Cruise	0,00	1,00
27	82	32.4.3	Cruise	0,00	1,00
28	216	32.4.3	Cruise	0,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	51	32.4.3	Cruise	0,00	1,00
30	277	32.4.3.5	Influence of variables on cruise performance	0,00	1,00
31	416	33.	FLIGHT PLANNING AND FLIGHT MONITORING	0,00	1,00
32	413	33.	FLIGHT PLANNING AND FLIGHT MONITORING	0,00	1,00
33	414	33.	FLIGHT PLANNING AND FLIGHT MONITORING	0,00	1,00
34	415	33.	FLIGHT PLANNING AND FLIGHT MONITORING	0,00	1,00
35	421	33.	FLIGHT PLANNING AND FLIGHT MONITORING	0,00	1,00
36	417	33.1	FLIGHT PLANNING FOR VFR FLIGHTS	0,00	1,00
37	226	33.1.1.1	Routes, airfields, heights and altitudes from VFR charts	0,00	1,00
38	259	33.1.1.5	Completion of navigation plan	0,00	2,00
39	30	33.5.1.1	Format of Flight Plan	0,00	1,00
40	203	33.5.1.1	Format of Flight Plan	0,00	1,00
Total:15%				6,00	41,00

Attachment 2: List of result by topic

Licence

Date: 17.07.2014

Examinee: Ariho, William

Location: Uganda Civil Aviation Authority

Your Result: 6,00 from 41,00 Points (14,63 %)

Topics	Amount of Questions	Points achieved	Maximum Points	Percent
31. MASS AND BALANCE - AEROPLANES/HELICOPTERS	15	5,00	15,00	33,33
31.2. LOADING	1	0,00	1,00	0,00
31.3. FUNDAMENTALS OF CG CALCULATIONS	1	1,00	1,00	100,00
32. PERFORMANCE - AEROPLANES	15	1,00	15,00	6,67
32.2. PERFORMANCE CLASS B - SINGLE-ENGINE AEROPLANES	2	0,00	2,00	0,00
32.3. PERFORMANCE CLASS B - MULTI-ENGINE AEROPLANES	1	0,00	1,00	0,00
32.4. PERFORMANCE CLASS A - AEROPLANES CERTIFICATED UNDER CS 25 ONLY	9	0,00	9,00	0,00
33. FLIGHT PLANNING AND FLIGHT MONITORING	10	0,00	11,00	0,00
33.1. FLIGHT PLANNING FOR VFR FLIGHTS	3	0,00	4,00	0,00
33.5. ICAO FLIGHT PLAN (ATS Flight Plan)	2	0,00	2,00	0,00
Total	40	6,00	41,00	14,63

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Date of birth: 1989-04-22

Subject: 030-Flight Performance&Planing
Date: 2014-07-17



1

What is the location of the CG if 146 pounds are removed from Station 150?

Aircraft weight 7,152 lb

CG location Station 82

☐ 80.6.

☐ 81.3.

☒ 83.4.

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2

What is the location of the CG if 60 pounds are removed from Station 70?

Aircraft weight 8,420 lb

CG location Station 85

☒ 85.1.

☐ 84.1.

☐ 84.9.

3

Could 100 pounds of weight be shifted from Station 30.0 to Station 120.0 without exceeding the aft CG limit?

Total weight 4,750 lb

CG location Station 115.8

☒ Yes; the CG would remain at Station 115.8.

☐ Yes; the new CG would be located at Station 117.69.

☐ No; the new CG would be located at Station 118.15.

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4

How much weight could be added at Station 120 without exceeding the aft CG limit?

Aircraft weight 9,500 lb

CG location Station 90.0

Aft CG limit Station 90.5

☐ 110.5 pounds.

☒ 61.0 pounds.

☐ 161.0 pounds.

5

The center of gravity of an aircraft can be determined by

- ☐ multiplying total arm by total weight.
- ☒ dividing total moment by total weight.
- ☐ dividing total arm by total moments.

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6

Under which condition is a forward CG most critical?

- ☒ On landing.
- ☐ When in an unusual attitude.
- ☒ On takeoff.

7

An aircraft is loaded with the CG at the aft limit. What are the performance characteristics compared with the CG at the forward limit?

- ☐ The aft CG provides the lowest stalling speed, the highest cruising speed, and least stability.
- ☐ The aft CG provides the highest stall speed and cruising speed.
- ☒ Cruising speed is lower because of more induced drag created by the elevator or stabilizer being required to provide more lift with an aft CG.

8

Could 100 pounds of weight be shifted from Station 130.0 to Station 30.0 without exceeding the forward CG limit?

Total weight 2,800 lb
CG location Station 120.0

- ☒ No; the new CG would be located at Station 116.42.
- ☐ Yes; the new CG would be located at Station 117.89.
- ☐ No; the new CG would be located at Station 116.89.

9

The center of gravity tube is

- ☒ shortened for lighter pilots.
- ☐ lengthened for lighter pilots.
- ☐ lengthened for heavier pilots.

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10

What is the maximum allowable weight that may be carried on a pallet which has the dimensions of 24.6 x 68.7 inches?
Floor load limit - 85 lb/sq ft
Pallet weight - 44 lb
Tiedown devices - 29 lb

☐ 924.5 pounds.

☒ 953.6 pounds.

☐ 968.6 pounds.

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11

How much weight must be shifted from Station 150.0 to Station 30.0 to move the CG to exactly the aft CG limit?

Total weight 7,500 lb

CG location Station 80.5

☐ 65.8 pounds.

☒ 68.9 pounds.

☐ 62.5 pounds.

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12

How much weight could be added at Station 160 without exceeding the aft CG limit?

Aircraft weight 8,300 lb
CG location Station 90.0
Aft CG limit Station 90.5

☒ 59.7 pounds.

☐ 13.9 pounds.

☐ 16.5 pounds.

13

The term 'weigh-off' as used in ballooning means to determine the

- ☐ static equilibrium of the balloon as loaded for flight.
- ☐ amount of fuel required for ascent to a preselected altitude.
- ☒ standard weight and balance of the balloon.

14

In flight the wing of an aircraft containing fuel is subjected to vertical loads that produce a bending moment which is:

- ☒ equal to the zero -fuel weight multiplied by the span
- ☐ highest at the wing root
- ☐ lowest at the wing root
- ☐ equal to half the weight of the aircraft multiplied by the semi span

15

When an aircraft is stationary on the ground, its total weight will act vertically

- ☐ *through a point defined as the datum point.*
- ☒ *through its centre of gravity.*
- ☐ *through the main wheels of its undercarriage assembly.*
- ☐ *through its centre of pressure.*

16

What is the best method of speed reduction if hydroplaning is experienced on landing?

- ☐ Apply full main wheel braking only.
- ☒ Apply aerodynamic braking to the fullest advantage.
- ☐ Apply nosewheel and main wheel braking alternately and abruptly.

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17

To level off from a descent maintaining the descending airspeed, the pilot should lead the desired altitude by approximately

☒ 20 feet.

☐ 60 feet.

☐ 50 feet.

18

To level off at an airspeed higher than the descent speed, the addition of power should be made, assuming a 500 FPM rate of descent, at approximately

- ☐ 100 to 150 feet above the desired altitude.
- ☐ 150 to 200 feet above the desired altitude.
- ☒ 50 to 100 feet above the desired altitude.

19

Which statement is correct for a descent without engine thrust at maximum lift to drag ratio speed?

- ☐ The higher the gross mass the lower is the speed for descent.
- ☒ The mass of an aeroplane does not have any effect on the speed for descent.
- ☐ The higher the average temperature (OAT) the lower is the speed for descent.
- ☐ The higher the gross mass the greater is the speed for descent.

20

The coefficient of lift can be increased either by flap extension or by

- ☐ increasing the CAS.
- ☐ increasing the TAS.
- ☐ decreasing the 'nose-up' elevator trim setting.
- ☒ increasing the angle of attack.

21

The decision speed at take-off (V1) is the calibrated airspeed:

- ☐ at which the take-off must be rejected.
- ☐ at which the failure of the critical engine is expected to occur.
- ☒ below which take-off must be rejected if an engine failure is recognized, above which take-off must be continued.
- ☐ below which the take-off must be continued.

22

Which of the following statements is correct?

- ☐ The accelerate stop distance required is independant of the runway condition.
- ☒ The climb limited take-off mass is independant of the wind component.
- ☐ The performance limited take-off mass is independant of the wind component.
- ☐ The take-off distance with one engine out is independant of the wind component.

23

What happens to the drag of a jet aeroplane if, during the initial climb after take off, constant IAS is maintained?
(Assume a constant mass.)

- ☐ The drag increases initially and decreases thereafter.
- ☐ The drag increases considerably.
- ☒ The drag remains almost constant.
- ☐ The drag decreases.

24

The climb limited take-off mass can be increased by

- ☒ a lower flap setting for take-off and selecting a higher V2.
- ☐ selecting a lower V2.
- ☐ selecting a lower V1.
- ☐ selecting a lower VR.

25

The best rate of climb at a constant gross mass

- ☐ increases with increasing altitude since the drag decreases due to the lower air density.
- ☐ is independent of altitude.
- ☒ decreases with increasing altitude since the thrust available decreases due to the lower air density.
- ☐ increases with increasing altitude due to the higher true airspeed.

26

The optimum cruise altitude is

- ☐ the pressure altitude up to which a cabin altitude of 8000 ft can be maintained.
- ☒ the pressure altitude at which the best specific range can be achieved.
- ☐ the pressure altitude at which the fuel flow is a maximum.
- ☐ the pressure altitude at which the speed for high speed buffet as TAS is a maximum.

27

Which of the following statements is correct?

- ☒ Induced drag decreases with increasing speed.
- ☐ Induced drag is independant of the speed.
- ☐ Induced drag decreases with increasing angle of attack.
- ☐ Induced drag increases with increasing speed.

28

Which force compensates the weight in unaccelerated straight and level flight ?

- ☐ the thrust
- ☐ the resultant from lift and drag
- ☐ the drag
- ☒ the lift

29

Which statement with respect to the step climb is correct?

- ☒ Executing a desired step climb at high altitude can be limited by buffet onset at g-loads larger than 1.
- ☐ A step climb is executed in principle when, just after leveling off, the 1.3g altitude is reached.
- ☐ A step climb is executed because ATC desires a higher altitude.
- ☐ A step climb must be executed immediately after the aeroplane has exceeded the optimum altitude.

30

Considering TAS for maximum range and maximum endurance, other factors remaining constant,

- ☐ *TAS for maximum range will increase with increased altitude while TAS for maximum endurance will decrease with increased altitude.*
- ☒ *both will increase with increasing altitude.*
- ☐ *both will stay constant regardless of altitude.*
- ☐ *both will decrease with increasing altitude.*

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31

If an aircraft is consuming 91 pounds of fuel per hour and groundspeed is 168 knots, how much fuel is required to travel 457 NM?

☐ 291 pounds.

☒ 248 pounds.

☐ 265 pounds.

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32

GIVEN:

Usable fuel at takeoff 40 gal, Fuel consumption rate 12.2 gal/hr Constant
groundspeed 120 kts
Flight time since takeoff 1 hr 30 min

☐ 216 NM.

☒ 121 NM.

☐ 156 NM.

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33

GIVEN:

Usable fuel at takeoff 40 gal
Fuel consumption rate 12.2 gal/hr
Constant groundspeed 120 kts
Flight time since takeoff 1 hr 30 min

☒ 176 NM.

☐ 215 NM.

☐ 121 NM.

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34

If fuel consumption is 15.3 gallons per hour and groundspeed is 167 knots, how much fuel is required for an aircraft to travel 620 NM?

☐ 60 gallons.

☒ 57 gallons.

☐ 63 gallons.

35

How is your flight plan closed when your destination airport has IFR conditions and there is no control tower or Flight control Center(FCC) on the field?

- ☐ The ARTCC controller will close your flight plan when you report the runway in sight.
- ☒ Upon landing, you must close your flight plan by radio or by telephone to any FCC or ATC facility.
- ☐ You may close your flight plan any time after starting the approach by contacting any FCC or ATC facility.

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36

How long will a Flight Service Station hold a VFR flight plan past the proposed departure time?

☐ 2 hours.

☒ 1 hour.

☐ 30 minutes.

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37

Given : true track 017; W/V 340/30; TAS 420 kt Find : wind correction angle (WCA) and ground speed (GS)

☐ WCA -2° ; GS 426 kt

☒ WCA -2° ; GS 396 kt

☐ WCA $+2^{\circ}$; GS 416 kt

☐ WCA $+2^{\circ}$; GS 396 kt

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38

Given: true course (TC) 017; W/V 340/30; TAS 420 kt
Find: wind correction angle (WCA) and ground speed (GS)

- ☐ WCA -2° ; GS 426 kt
- ☐ WCA $+2^{\circ}$; GS 396 kt
- ☒ WCA -2° ; GS 396 kt
- ☐ WCA $+2^{\circ}$; GS 416 kt

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39

If your destination airport has no ICAO indicator, in the appropriate box of your flight plan, you write:

☐ A AAAA

☐ C IIII

☒ D ZZZZ

☐ B XXXX

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40

When completing an ATC flight plan, an elapsed time (Item 16) of 1 hour 55 minutes should be entered as:

☐ 1H55

☐ 0115

☒ 0155

☐ 115M